

1. A modern CPU improves performance mainly by:

- a. Increasing clock speed only
- b. Using multiple cores
- c. Removing cache
- d. Using larger RAM

Answer : B

2. Why is parallel computing used?

- a. To solve large/complex problems faster
- b. To increase memory size
- c. To reduce programming complexity
- d. To execute tasks one by one

Answer : A

3. SIMD is ideal for data-level parallelism.

Pilih salah satu:

Benar

Salah

Answer : Benar

4. Which architecture executes a single instruction on multiple data?

- a. SSID
- b. MISD
- c. MIMD
- d. SIMD

Answer : D

5. What is the main goal of parallel computing?

- a. Reduce memory usage
- b. Increase clock frequency
- c. Execute multiple tasks simultaneously
- d. Replace sequential programming

Answer : C

6. Serial computing means:

- a. Many cores execute tasks
- b. Execution on GPU
- c. Tasks executed simultaneously
- d. Execution on a single core

Answer : D

7. MIMD processors execute instructions asynchronously.

Pilih salah satu:

Benar

Salah

Answer : Benar

8. In seial computing, tasks run one after another.

Pilih salah satu:

Benar

Salah

Answer: Benar

9. Parallel computing always guarantees faster execution. Pilih salah satu:

Benar

Salah

Answer : Salah

10. Parallel computing works by:

- a. Disabling threads
- b. Splitting tasks across cores
- c. Running only one instruction
- d. Increasing task's size

Answer :B

Pertemuan Kedua!

11. What is the core idea of parallel programming?

- a. Running a program faster on a single, powerful processor.
- b. Dividing a problem into smaller parts that execute simultaneously on multiple processing units.
- c. Writing code that is longer and more complex to improve accuracy.
- d. Using a single thread to manage all tasks sequentially.

Answer : b

12. A race condition is a desired outcome in parallel programming where threads compete to finish a task first.

Pilih salah satu:

Benar

Salah

Answer : salah

13. What is the key difference between threads in the same process?

- a. They must always run the exact same code.
- b. They share the same memory space but have their own execution path.
- c. They do not share any memory or resources.
- d. One thread controls all the others.

Answer : B

14. Which of the following are the first steps in creating a parallel program?

- a. Partitioning the work and its associated data.
- b. Writing the entire program as a single, sequential block of code first.
- c. Managing data access, communication, and synchronization.
- d. Identifying work that can be performed in parallel.

Answer : A, C AND D

15. Amdahl's Law explains that the potential speedup of a program is limited by the portion of the program that cannot be parallelized.

Pilih salah satu:

Benar

Salah

Answer : Benar

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16. In the context of parallel programming, what is a thread?

- a. A type of memory used only for I/O operations.
- b. The smallest unit of execution within a process that can run concurrently.
- c. A single, complete program that runs independently.
- d. A physical cable connecting different processing units.

Answer : B

17. A problem that consists of large, independent datasets is generally a bad candidate for parallelization.

Pilih salah satu:

Benar

Salah

Answer : salah

Latihan Soal

18. Definition of Parallel Computing

English:

What is parallel computing?

- A. Executing multiple tasks sequentially on one processor
- B. Executing multiple tasks simultaneously using multiple processors
- C. Running a program without memory
- D. Processing data manually

Answer : B

Indonesian:

Apa yang dimaksud dengan komputasi paralel?

- A. Menjalankan banyak tugas secara berurutan pada satu prosesor
- B. Menjalankan banyak tugas secara bersamaan menggunakan beberapa prosesor
- C. Menjalankan program tanpa memori
- D. Memproses data secara manual

19. Purpose of Parallel Computing

English:

Why is parallel computing used?

- A. To slow down computation
- B. To reduce hardware usage
- C. To solve complex problems faster
- D. To eliminate processors

Answer : C

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Indonesian:

Mengapa komputasi paralel digunakan?

- A. Untuk memperlambat komputasi
- B. Untuk mengurangi penggunaan perangkat keras
- C. Untuk menyelesaikan masalah kompleks lebih cepat
- D. Untuk menghilangkan prosesor

20. Flynn's Taxonomy

English:

Which architecture executes a single instruction on multiple data?

- A. SISD
- B. SIMD
- C. MISD
- D. MIMD

Answer : B

Indonesian:

Arsitektur mana yang menjalankan satu instruksi pada banyak data?

- A. SISD
- B. SIMD
- C. MISD
- D. MIMD

21. SISD Architecture

English:

Which of the following is an example of SISD?

- A. GPU
- B. Multicore CPU
- C. Intel 8086
- D. Supercomputer

Indonesian:

Manakah contoh arsitektur SISD?

- A. GPU
- B. CPU multicore
- C. Intel 8086
- D. Superkomputer

Answer : C

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22. MISD Characteristic

English:

Which architecture is rarely used and mainly theoretical?

- A. SISD
- B. SIMD
- C. MISD
- D. MIMD

Indonesian:

Arsitektur mana yang jarang digunakan dan bersifat teoritis?

- A. SISD
- B. SIMD
- C. MISD
- D. MIMD

Answer : C

23. MIMD Architecture

English:

Which architecture is commonly used in modern computing systems?

- A. SISD
- B. SIMD
- C. MISD
- D. MIMD

Answer : D

Indonesian:

Arsitektur mana yang paling umum digunakan pada sistem modern?

- A. SISD
- B. SIMD
- C. MISD
- D. MIMD

24. Parallel Programming

English:

Parallel programming involves:

- A. Executing one instruction at a time
- B. Running tasks simultaneously
- C. Ignoring data dependencies
- D. Using only one processor

Answer :B

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Indonesian:

Pemrograman paralel melibatkan:

- A. Menjalankan satu instruksi pada satu waktu
- B. Menjalankan tugas secara bersamaan
- C. Mengabaikan ketergantungan data
- D. Menggunakan satu prosesor saja

25. Data Parallelism

English:

Data parallelism refers to:

- A. Different operations on the same data
- B. Same operation on different data
- C. Sequential execution
- D. Single-thread execution

Indonesian:

Data parallelism mengacu pada:

- A. Operasi berbeda pada data yang sama
- B. Operasi yang sama pada data yang berbeda
- C. Eksekusi berurutan
- D. Eksekusi satu thread

Answer : B

26. Task Parallelism

English:

Task parallelism involves:

- A. One task on one processor
- B. Same tasks on different data
- C. Different tasks executed simultaneously
- D. Serial processing

Answer : C

Indonesian:

Task parallelism melibatkan:

- A. Satu tugas pada satu prosesor
- B. Tugas yang sama pada data berbeda
- C. Tugas berbeda yang dijalankan secara bersamaan
- D. Pemrosesan serial

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27. Thread Definition

English:

A thread is:

- A. A complete program
- B. The smallest unit of execution within a process
- C. A type of processor
- D. A storage device

Indonesian:

Thread adalah:

- A. Sebuah program lengkap
- B. Unit eksekusi terkecil dalam suatu proses
- C. Jenis prosesor
- D. Perangkat penyimpanan

Answer : B

28. Amdahl's Law

English:

Amdahl's Law describes:

- A. Memory allocation
- B. Limits of parallel speedup
- C. CPU temperature
- D. Network communication

Indonesian:

Hukum Amdahl menjelaskan:

- A. Alokasi memori
- B. Batas percepatan komputasi paralel
- C. Suhu CPU
- D. Komunikasi jaringan

Answer : B

29. Synchronization

English:

Synchronization is used to:

- A. Increase program errors
- B. Prevent data inconsistency
- C. Slow down execution
- D. Disable processors

Answer : B

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Indonesian:

Sinkronisasi digunakan untuk:

- A. Meningkatkan kesalahan program
- B. Mencegah inkonsistensi data
- C. Memperlambat eksekusi
- D. Menonaktifkan prosesor

30. Race Condition

English:

A race condition occurs when:

- A. Only one thread accesses data
- B. Threads access shared data without synchronization
- C. Programs run sequentially
- D. Data is encrypted

Indonesian:

Race condition terjadi ketika:

- A. Hanya satu thread mengakses data
- B. Thread mengakses data bersama tanpa sinkronisasi
- C. Program berjalan secara berurutan
- D. Data dienkripsi

Answer : B

31. Deadlock

English:

Deadlock occurs when:

- A. A process finishes early
- B. Two or more processes wait indefinitely for each other
- C. A program runs successfully
- D. Memory increases

Indonesian:

Deadlock terjadi ketika:

- A. Sebuah proses selesai lebih awal
- B. Dua atau lebih proses saling menunggu tanpa batas
- C. Program berjalan dengan baik
- D. Memori meningkat

Answer : B

32. Load Balancing

English:

Load balancing aims to:

- A. Overload one processor

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- B. Distribute tasks evenly across processors
- C. Reduce memory usage
- D. Stop parallel execution

Answer :B

Indonesian:

Load balancing bertujuan untuk:

- A. Membebani satu prosesor
- B. Mendistribusikan tugas secara merata
- C. Mengurangi penggunaan memori
- D. Menghentikan eksekusi parallel

33. Static Load Balancing

English:

Static load balancing assigns tasks:

- A. During execution
- B. Before execution begins
- C. Randomly
- D. After execution ends

Answer :B

Indonesian:

Static load balancing membagi tugas:

- A. Saat eksekusi berlangsung
- B. Sebelum eksekusi dimulai
- C. Secara acak
- D. Setelah eksekusi selesai

34. Dynamic Load Balancing

English:

Dynamic load balancing distributes tasks:

- A. Before execution only
- B. During execution
- C. Sequentially
- D. Manually

Answer :B

Indonesian:

Dynamic load balancing mendistribusikan tugas:

- A. Hanya sebelum eksekusi
- B. Selama eksekusi

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- C. Secara berurutan
- D. Secara manual

35. Communication Overhead

English:

Communication overhead refers to:

- A. Computation time only
- B. Extra time for coordination and data exchange
- C. Hardware installation time
- D. Idle processor time only

Answer :B

Indonesian:

Communication overhead adalah:

- A. Waktu komputasi saja
- B. Waktu tambahan untuk koordinasi dan pertukaran data
- C. Waktu pemasangan perangkat keras
- D. Waktu prosesor menganggur saja

36. OpenMP

English:

OpenMP is used for:

- A. Distributed memory systems
- B. Shared memory parallel programming
- C. Sequential programming
- D. Networking only

Answer : B

Indonesian:

OpenMP digunakan untuk:

- A. Sistem memori terdistribusi
- B. Pemrograman paralel memori bersama
- C. Pemrograman sekuensial
- D. Jaringan saja

37. MPI

English:

MPI is designed for:

- A. Single-core processors
- B. Distributed memory systems

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- C. Serial computing
- D. Local storage management

Answer :B

Indonesian:

MPI dirancang untuk:

- A. Prosesor inti tunggal
- B. Sistem memori terdistribusi
- C. Komputasi serial
- D. Manajemen penyimpanan lokal

38. Hybrid Model

English:

The hybrid parallel model combines:

- A. SISD and SIMD
- B. OpenMP and MPI
- C. Serial and sequential computing
- D. CPU and RAM

Indonesian:

Model paralel hybrid menggabungkan:

- A. SISD dan SIMD
- B. OpenMP dan MPI
- C. Komputasi serial dan sekuensial
- D. CPU dan RAM

Answer :B

39. For-loop pada Python pada umumnya akan lebih lambat karena Python merupakan salah satu interpreted language.

Answer : True

40. Single Instruction, Multiple Data (SIMD) programming digunakan pada:

- a. Distributed System
- b. Parallel Loop
- c. Vectorization
- d. Functional Programming

Answer : C

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41. Pada pemrograman Graphical Processing Unit (GPU), parallel paradigm yang umum digunakan adalah:

- a. Functional Programming
- b. Distributed System
- c. Parallel Loop
- d. Vectorization

Answer : D

42. Pada functional programming, dengan menggunakan konsep multiprocessing, perlu terdapat messaging antar process.

- a. Benar
- b. Salah

Answer : True

43. Paradigma yang digunakan adalah:

```
@jit(parallel=True)
def simulator(out):
    # iterate loop in parallel
    for i in prange(out.shape[0]):
        out[i] = run_sim()
```

- a. Paralel Loop
- b. Vectorization
- c. Distributed System
- d. Functional Programming

Answer : A

44. Benefit pemrograman parallel akan bergantung pada problem size.

- a. Benar
- b. Salah

= Answer :A

45. Penerapan parallelism para algoritma machine learning k-NN dapat menggunakan vektorisasi.

- a. Benar
- b. Salah

Answer : a

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46. Manakah dibawah ini yang bukan merupakan karakteristik dari komputasi paralel?

- a. Simultaneous execution
- b. Memecah task menjadi beberapa bagian
- c. Single processor
- d. Dapat memanfaatkan jaringan

Answer : C

47. Manakah yang BUKAN termasuk paradigma pemrograman paralel?

- a. Vectorization
- b. Distributed System
- c. Parallel Loop
- d. Functional Programming

Answer : B

48. Berakhirnya era “The Free Lunch” dalam sejarah perkembangan mikroproessor ditandai dengan:

- a. Semua benar
- b. Penambahan jumlah core pada suatu mikroporsesor
- c. Saturasi clock-speed pada suatu mikroprosesor
- d. Saturasi pada power consumption suatu mikroporsor

Answer : A

49. Pada pemrograman dengan multiprocessing, parallel paradigm yang umum digunakan adalah:

- a. Distributed System
- b. Vectorization
- c. Parallel Loop
- d. Functional Programming

Answer : D

50. Pada functional programming, dengan menggunakan konsep multiprocessing, perlu terdapat messaging antar process.

- a. Salah
- b. Benar

Answer : B

51. Pada pemrograman Graphical Processing Unit (GPU), parallel paradigm yang umum digunakan adalah:

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- a. Parallel Loop
- b. Distributed System
- c. Vectorization
- d. Functional Programming\

Answer : C

52. Penggunaan "for" dan "prange" pada Python dengan NUMBA termasuk:

- a. Parallel Loop
- b. Functional Programming
- c. Distributed System
- d. Vectorization

Answer : A

53. Penerapan parallelism pada algoritma machine learning k-NN dapat menggunakan vektorisasi.

- a. Benar
- b. Salah

Answer : A

54. State flattening merupakan cara untuk melakukan paralelisasi data dan umumnya terjadi pada saat melakukan vektorisasi.

- a. Salah
- b. Benar

Answer : B

55. Single Instruction, Multiple Data (SIMD) programming digunakan pada:

- a. Vectorization
- b. Distributed System
- c. Functional Programming
- d. Parallel Loop

Answer : A

56. Teknik dalam arsitektur suatu prosesor yang memungkinkan satu instruksi untuk memproses banyak data secara bersamaan disebut sebagai:

- a. SIMD
- b. MMX
- c. ISA
- d. MIMD

Answer : A

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57.

Paradigma apakah yang direpresentasikan ilustrasi di bawah ini :

☐ a. Parallel Loop

☐ b. Functional Programming

☐ c. Multiprocessing

☐ d. Vektorisasi

Answer : A